

Microeconomics I

Class 2 — Monotonicity, convexity, indifference curves

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slide 1 of 33

Slide 1 Title

Class 2 covers one section of the textbook, MWG 3.B: the assumptions economists place on preferences over bundles of goods, and what each assumption does to the picture of indifference curves.

Plan for today

10:00	Sign-up sheet; three snacks
10:10	Warm-up: two relations
10:20	Quiz results: $\geq$ vs $\gg$, $\forall$ and $\exists$
10:30	Monotonicity, local non-satiation
11:00	Break
11:10	Indifference and contour sets
11:20	Convexity, the MRS
11:35	Five preferences, in pairs
11:50	Homothetic, quasilinear; summary

Slide 2 Plan for today

The morning at a glance. The first half sharpens tools: relations, vector inequalities, quantifiers, monotonicity. The second half draws the pictures: contour sets, convexity, five worked preferences. Everything today is MWG section 3.B; Muñoz-Garcia chapter 1 covers the same ground more gently.

Practicalities

- Group sign-up sheet going round: 2–5 names per group
- Problem set 1 out Tue 6 Oct, due in class Wed 14 Oct
- Office hours: today 12:00–13:00, after class
up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Slide 3 Practicalities

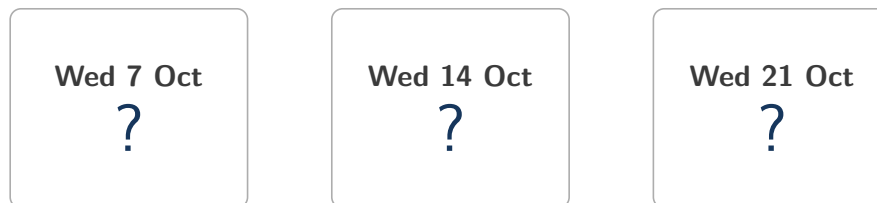
Problem sets are done in groups of 2 to 5; one line per group on the sign-up sheet.

Problem set 1 is handed out on Tuesday 6 October and is due in class on Wednesday 14 October, one paper copy per group.

Office hours: after class, up to 20 minutes per student; booking by e-mail gives priority.

Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu.

Three classes, three snacks



menu: chocolate bar · crisps · nuts · biscuits · fruit · wafer

Choose all three now. Hands up: three different ones?

Set-up after Simonson (1990), study 1.

Slide 4 Three classes, three snacks

A thought experiment to open. You must order a snack for each of the next three Wednesday classes, all three now, from a menu of six. Would you pick three different ones?

Keep your answer in mind. The next slide gives the data from a real version of this experiment; slide 26 gives the economics.

Chosen as a bundle, people mix. One at a time, they repeat.

all three chosen today

64 %

took three different snacks

one chosen each week

9 %

took three different snacks

Same menu, students in a class. Which of the two choices reveals the **preference**?

Simonson (1990), *Journal of Marketing Research* 27(2), study 1. We come back to it once “convex” is defined.

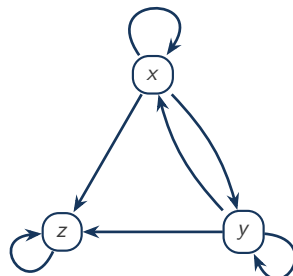
Slide 5 Chosen as a bundle, people mix

Simonson (1990) ran this with students in a class. Those who chose all three snacks at once: 64 % took three different ones. Those who chose one snack each week, just before eating it: 9 % ended up with three different ones. Same menu, same kind of people, different framing of the choice.

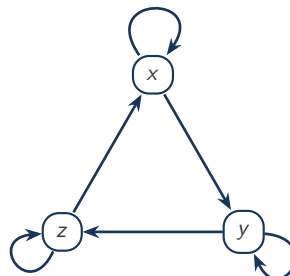
The question is left open on purpose: which of the two choices is “the preference”? We come back to it once convexity is defined (slide 26).

Caveat: two groups of subjects and aggregate shares, not the same person choosing twice.

In pairs: two relations on $\{x, y, z\}$



R_1



R_2

$a \rightarrow b$ means $a R b$. Complete, reflexive, transitive, symmetric, asymmetric, antisymmetric?

Four minutes. Which one is a **rational** preference?

Slide 6 In pairs: two relations

Two relations on three objects, drawn as graphs: an arrow from a to b means $a R b$, and every node has a loop ($a R a$).

Try it: for each relation decide the six properties from class 1 (complete, reflexive, transitive, symmetric, asymmetric, antisymmetric), then whether it is a rational preference, that is complete and transitive.

It helps to list the pairs first. R_2 is $x R y, y R z, z R x$ plus the three loops.

Answers: two relations

	R_1	R_2
complete	✓	✓ every pair one way, all loops
reflexive	✓	✓ three loops
transitive	✓	✗ $x R y, y R z$, but not $x R z$
symmetric	✗ $x R z$, not $z R x$	✗ $x R y$, not $y R x$
asymmetric	✗ loops; $x R y$ and $y R x$	✗ loops
antisymmetric	✗ $x R y, y R x, x \neq y$	✓ never both ways

R_1 is rational: $x \sim y \succ z$. R_2 is complete but cycles, $x \succ y \succ z \succ x$: the money pump.

Slide 7 Answers: two relations

R_1 is complete and transitive, so it is a rational preference: x and y are indifferent, and both are strictly preferred to z .

R_2 is complete too: every pair is related in at least one direction, and the loops give $x R x$. But it is not transitive: $x R y$ and $y R z$, yet the arrow between x and z runs the other way. That is the cycle $x \succ y \succ z \succ x$, the money pump from class 1.

Lesson: completeness and transitivity are different axioms, and it is transitivity that rules out cycles. Asymmetry and antisymmetry only look at pairs, so a cycle can pass antisymmetry, as R_2 does, and still be irrational.

From class 1: what today builds on

$\succsim$ is **rational**: complete and transitive

MWG Def. 1.B.1

$x \succ y \iff x \succsim y$ and not $y \succsim x$ $x \sim y \iff x \succsim y$ and $y \succsim x$

$\sim$ is an equivalence relation: X splits into **indifference classes**

Prop. 1.B.1

Today: what those classes look like once $\succsim$ is **monotone** and **convex**.

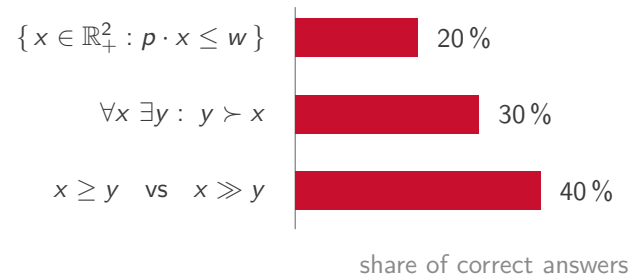
Slide 8 From class 1

The three facts from class 1 that today rests on. A preference relation $\succsim$ is rational if it is complete (any two bundles can be compared) and transitive (no cycles). Strict preference $x \succ y$ means x is at least as good as y but not the other way round; indifference $x \sim y$ means each is at least as good as the other.

Because $\sim$ is an equivalence relation, the consumption set X splits into indifference classes.

Today's question: what shape do those classes have once we add monotonicity and convexity? With two goods they become the familiar indifference curves.

Diagnostic quiz: three things to fix today



$n = 20$, mean 3.9 of 8. The ideas were fine (tangency: 95%); the **notation** was not.

Slide 9 Diagnostic quiz

The diagnostic quiz from class 1: 20 answers, mean 3.9 of 8. The economics was fine (95 % got the tangency question). Three pieces of notation were not: set-builder notation for the budget set (20 % correct), the order of the quantifiers $\forall$ and $\exists$ (30 %), and $\geq$ versus $\gg$ for vectors (40 %).

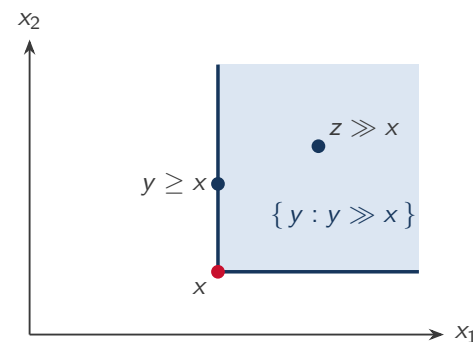
All three reappear within the hour, so the next two slides fix them; set-builder notation returns on slide 21.

Two ways to say “more”

$$y \geq x \iff y_1 \geq x_1 \text{ and } y_2 \geq x_2$$

$$y \gg x \iff y_1 > x_1 \text{ and } y_2 > x_2$$

Quiz: $x = (3, 1)$, $y = (3, 2)$.



Shaded: strictly more of both. Its edges belong to $\geq$ but not to $\gg$.

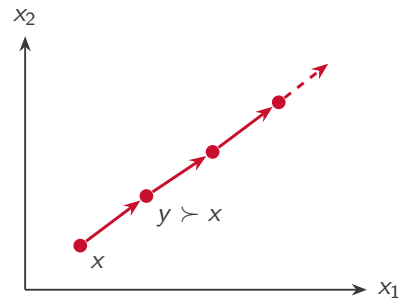
Slide 10 Two ways to say “more”

Two inequalities between bundles. $y \geq x$ means y has at least as much of every good as x ; a tie in some good is allowed. $y \gg x$ means y has strictly more of every good.

In the picture the shaded open quadrant is the set of bundles $\gg x$. Its two edges (same amount of one good, more of the other) satisfy $\geq$ but not $\gg$. Quiz example: $y = (3, 2)$ versus $x = (3, 1)$: $y \geq x$ holds, $y \gg x$ fails because good 1 ties.

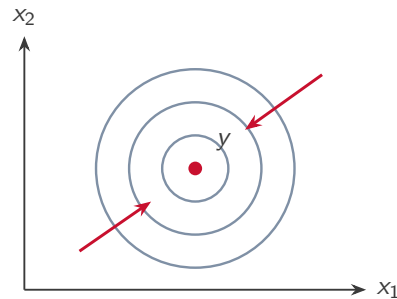
The combination “ $y \geq x$ and $y \neq x$ ” (at least as much of both, strictly more of at least one) is exactly what strong monotonicity uses on slide 13.

Swap the quantifiers, change the claim



$$\forall x \exists y : y \succ x$$

for every bundle, a better one



$$\exists y \forall x \neq y : y \succ x$$

one bundle beats all: a bliss point

Left: no bundle is best — **non-satiation**. Its local version arrives in ten minutes.

Slide 11 Swap the quantifiers

Same symbols, opposite claims. Left: for every bundle x there is a y with $y \succ x$. No bundle is best; wherever you are, something beats it (the chain of arrows). Right: there is one y that beats every other bundle, a bliss point; the circles around it are its indifference curves.

Swapping $\forall$ and $\exists$ is the whole difference between “no best bundle” and “a best bundle”.

The left statement is non-satiation. Its local version, slide 15, reads “for all x , for all ε , there exists y ”.

Slide 12 Monotonicity

Part 1: three versions of “more is better” (MWG 3.B), and the weakest one that still gives thin indifference curves.

Monotonicity

MWG 3.B

Definition: monotone, strongly monotone

$\succsim$ on $X = \mathbb{R}_+^2$ is **monotone** if

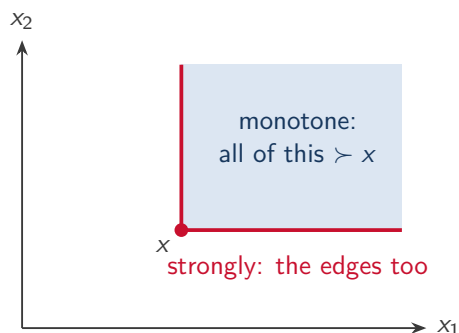
$$y \gg x \Rightarrow y \succ x$$

strongly monotone if

$$y \geq x \text{ and } y \neq x \Rightarrow y \succ x$$

MWG Definition 3.B.2.

Strongly monotone $\Rightarrow$ monotone. The converse fails — an example in the exercise.



Slide 13 Definition: monotone, strongly monotone

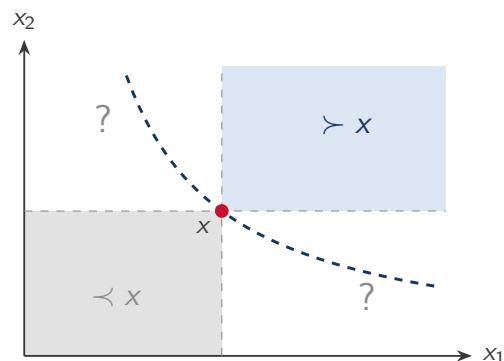
MWG Definition 3.B.2, on the consumption set $X = \mathbb{R}_+^2$ (two goods, non-negative amounts). $\succsim$ is monotone if a bundle with strictly more of every good is strictly preferred: $y \gg x$ implies $y \succ x$. It is strongly monotone if a bundle with at least as much of every good and more of at least one is already strictly preferred: $y \geq x$ and $y \neq x$ imply $y \succ x$.

Picture: monotonicity claims the open shaded quadrant above and to the right of x ; strong monotonicity also claims its two edges.

Strongly monotone implies monotone, because $y \gg x$ is a special case of $y \geq x$, $y \neq x$. The converse fails: perfect complements, in Exercise 1 (slide 18).

Slide 14 Where can the indifferent bundles be?

Monotone: where can the indifferent bundles be?



Nothing indifferent to x lies strictly north-east or strictly south-west of it. The curve through x cannot slope up.

What monotonicity does to the indifference curve through x . Everything strictly north-east of x is strictly better (blue). Everything strictly south-west is strictly worse (grey), because there $x \gg y$.

So a bundle indifferent to x can only lie in the two remaining quadrants or on their edges: the curve through x cannot slope upward.

Monotonicity alone still allows flat or vertical stretches along the edges, as in the L-shaped curves of perfect complements. Strong monotonicity removes them, and the curve is strictly downward sloping.

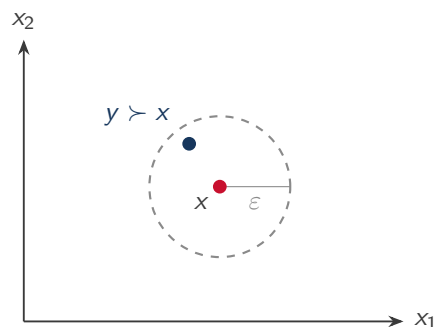
Definition: local non-satiation

$\succsim$ is **locally non-satiated** if
for every $x \in X$ and every $\varepsilon > 0$
there is $y \in X$ with

$$\|y - x\| \leq \varepsilon \quad \text{and} \quad y \succ x$$

MWG Definition 3.B.3.

$\forall x \forall \varepsilon \exists y$: the quiz's quantifier order. No direction is prescribed: y may have **less** of a good.



Slide 15 Definition: local non-satiation

MWG Definition 3.B.3. $\succsim$ is locally non-satiated if for every bundle x and every distance $\varepsilon > 0$, however small, there is a bundle y within distance ε of x that is strictly better than x . Read the quantifiers in order: for all x , for all ε , there exists y .

No direction is prescribed. In the picture y has less of good 1 and more of good 2. So local non-satiation allows “bads” such as garbage or pollution, as long as something nearby is better.

Where does it fail for the bliss-point preference of slide 11? At the bliss point, and only there. One bundle is enough to fail it, because the definition says “for every x ”.

Three strengths of “more is better”



Prove monotone $\Rightarrow$ locally non-satiated in two lines. Where do you use $X = \mathbb{R}_+^2$?

Grey: a preference with the weaker property but not the stronger one, written as the number it ranks by.

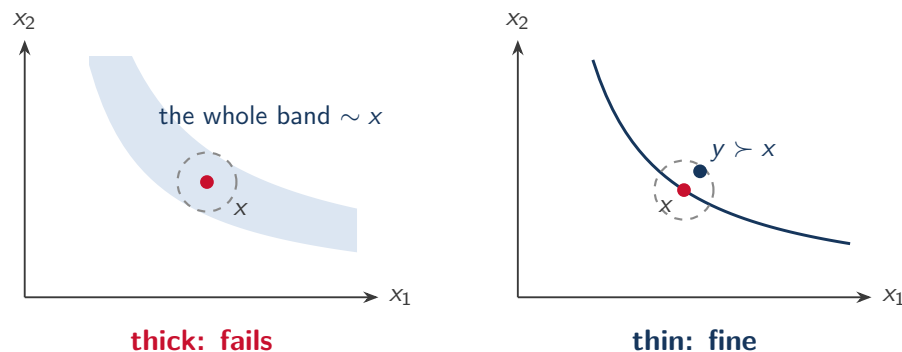
Slide 16 Three strengths of “more is better”

Strongly monotone implies monotone, which implies locally non-satiated; neither arrow reverses. The grey formulas are the counterexamples, each written as the number the preference ranks bundles by: $\min\{x_1, x_2\}$ (perfect complements) is monotone but not strongly monotone; $x_1 - x_2$ (good 2 is a bad) is locally non-satiated but not monotone, since $(2, 3)$ has more of both than $(1, 1)$ yet ranks lower.

Proof that monotone implies locally non-satiated: given x and ε , take $y = x + (\varepsilon/2, \varepsilon/2)$. Then $y \gg x$, so $y \succ x$, and the distance from x to y is $\varepsilon/\sqrt{2} < \varepsilon$.

Where is $X = \mathbb{R}_+^2$ used? To make sure y is an available bundle. On a bounded consumption set (say, at most 24 hours of leisure) the top corner has no such y , and monotone does not give local non-satiation there.

Local non-satiation rules out thick indifference sets



Inside a thick band, every bundle within ε of x is indifferent to x .

After MWG Figure 3.B.2, redrawn; the circle test on the right is MWG Figure 3.B.1.

Slide 17 Thick indifference sets

Why the book bothers with local non-satiation. Left: a thick band of mutually indifferent bundles. From a bundle x inside the band, every bundle within a small radius is indifferent to x , so no strictly better bundle exists nearby: local non-satiation fails. Right: a thin curve; any circle around x reaches the better side.

Local non-satiation is the weakest of the three assumptions that rules out thick indifference sets, and it is all that Walras' law will need later in the course. After MWG Figure 3.B.2; the circle test is MWG Figure 3.B.1.

In pairs: monotone? strongly? locally non-satiated?

$x \succsim y \iff$	monotone	strongly monotone	LNS
$x_1 + x_2 \geq y_1 + y_2$	✓	✓	✓
$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$	✓	✗	✓
$x_1 - x_2 \geq y_1 - y_2$	✗	✗	✓
$\ x - (5, 5)\ \leq \ y - (5, 5)\ $	✗	✗	✗
$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$	✓	✓	✓

Five minutes. Give a counterexample for every ✗.

min: $(1, 1) \sim (2, 1)$. $x_1 - x_2$: $(2, 3) \prec (1, 1)$. Bliss: nothing beats $(5, 5)$.

Slide 18 In pairs: monotone? strongly? LNS?

Five preferences, each given by the rule for “ x is at least as good as y ”. Try each row before reading the ticks; find a counterexample for every cross. None of the five has thick indifference sets.

(1) Perfect substitutes: any increase in either good raises the sum; all three hold.

(2) Perfect complements: more of both raises the minimum, so monotone, hence LNS. But $(1, 1) \sim (2, 1)$ (same minimum), so not strongly monotone.

(3) $x_1 - x_2$: good 2 is a bad. $(2, 3) \gg (1, 1)$ yet ranks lower: neither monotone nor strongly monotone. A small step to the right is always better: LNS holds.

(4) Bliss point $(5, 5)$: $(6, 6) \gg (5, 5)$ but is worse; at $(5, 5)$ nothing nearby is better, so LNS fails at one bundle, which is enough.

(5) Lexicographic: if $y \geq x$ and $y \neq x$, then $y_1 > x_1$, or $y_1 = x_1$ and $y_2 > x_2$; either way $y \succ x$: strongly monotone, hence the rest.

Slide 19 Break

Ten minutes.

Break

Back at 11:10

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Slide 20 Indifference and contour sets

Part 2: the sets an indifference curve is made of, and why curves do not cross. MWG 3.B.

Indifference and contour sets

MWG 3.B

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Three sets attached to a bundle x

$$\{y \in X : y \succeq x\}$$

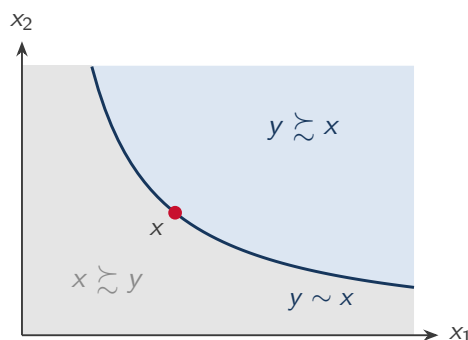
all bundles y in X such that y is at least as good as x

upper contour set $\{y \in X : y \succeq x\}$

indifference set $\{y \in X : y \sim x\}$

lower contour set $\{y \in X : x \succeq y\}$

MWG 3.B, p. 43, in the text; Figure 3.B.2(b).



Slide 21 Three sets attached to a bundle

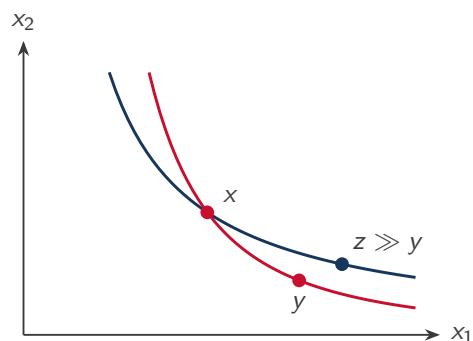
MWG 3.B, p. 43 (defined in the text, drawn in Figure 3.B.2(b)), and the notation from the quiz. Read $\{y \in X : y \succeq x\}$ as “the set of all bundles y in X such that y is at least as good as x ”: the braces build a set, the colon reads “such that”, the condition selects. The budget set $\{x \in \mathbb{R}_+^2 : p \cdot x \leq w\}$ uses the same grammar.

Three sets attached to a bundle x : the upper contour set, all bundles at least as good as x (blue, on and above the curve); the indifference set, all bundles indifferent to x (the curve); the lower contour set, all bundles that x is at least as good as (grey, on and below). Note that x belongs to all three and the curve belongs to both contour sets.

Which of the three is a budget set most like? A lower contour set: rank bundles by what they cost (dearer ranks higher); the budget set is everything ranked at or below a bundle costing exactly w .

Slide 22 Indifference curves do not cross

Indifference curves do not cross



$$y \sim x \text{ and } x \sim z \Rightarrow y \sim z$$

transitivity

$$z \gg y \Rightarrow z \succ y$$

monotone

Contradiction.

Rational + monotone: one indifference curve through every bundle; further out is better.

Why two indifference curves cannot cross when $\succeq$ is rational and monotone. Suppose they cross at x . Pick y on the red curve and z on the blue curve with z strictly north-east of y , so $z \gg y$. Since $y \sim x$ and $x \sim z$, transitivity gives $y \sim z$. But $z \gg y$, so monotonicity gives $z \succ y$. Contradiction.

Each axiom does one job: transitivity forbids the crossing, monotonicity orders the curves. So exactly one indifference curve passes through each bundle, and curves further from the origin are better.

Convexity

MWG 3.B

slide 23 of 33

Slide 23 Convexity

Part 3: convexity, a taste for mixing, and the marginal rate of substitution. MWG 3.B.

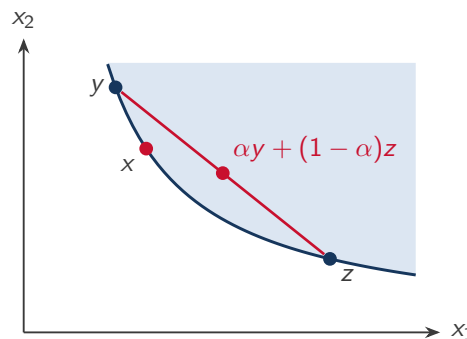
Definition: convex preferences

$\succsim$ is **convex** if every upper contour set $\{y \in X : y \succsim x\}$ is a convex set:

$$y \succsim x \text{ and } z \succsim x \\ \Rightarrow \alpha y + (1 - \alpha)z \succsim x, \quad \alpha \in [0, 1]$$

MWG Definition 3.B.4.

Averages are at least as good as extremes. The curve bows toward the origin.



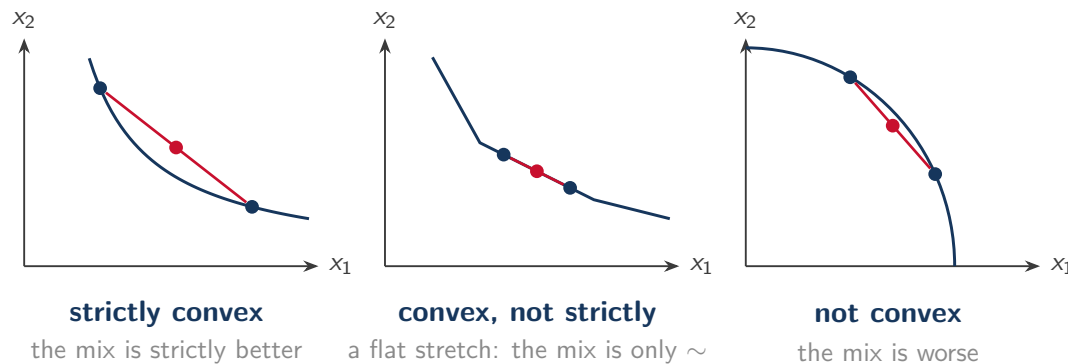
Slide 24 Definition: convex preferences

MWG Definition 3.B.4. $\succsim$ is convex if every upper contour set is a convex set: whenever y and z are both at least as good as x , so is every weighted average $\alpha y + (1 - \alpha)z$ with α between 0 and 1. In the picture y and z lie on the curve through x , and the red chord between them stays inside the blue set.

In words: averages are at least as good as extremes, and the indifference curve bows toward the origin. This is the convex set of class 1 (the segment between any two points of the set stays in the set), applied to the upper contour set of every bundle.

Asking instead that the lower contour set be convex is a different property: curves bowed away from the origin, the right panel of the next slide.

Strictly convex, convex, not convex



strictly convex: $y \succsim x, z \succsim x, y \neq z \Rightarrow \alpha y + (1 - \alpha)z \succ x, \alpha \in (0, 1)$

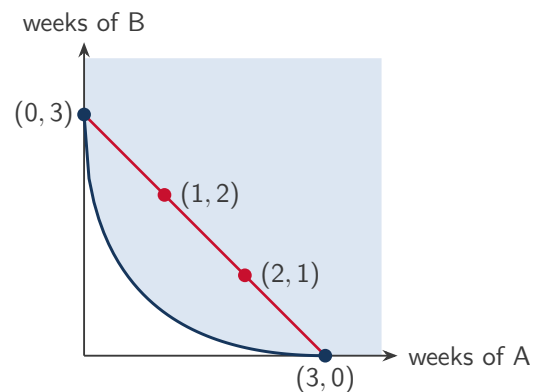
MWG Definition 3.B.5. Blue curve: an indifference set; blue dots: two bundles on it; red dot: their mix.

Slide 25 Strictly convex, convex, not convex

MWG Definition 3.B.5. $\succsim$ is strictly convex if, whenever y and z are two different bundles both at least as good as x , every mix $\alpha y + (1 - \alpha)z$ with $0 < \alpha < 1$ is strictly better than x .

Three panels, each with two blue bundles on an indifference curve and their mix in red. Strictly convex: the chord lies strictly above the curve, so the mix is strictly better. Convex but not strictly: along a linear stretch the chord stays on the curve, so the mix is only indifferent; this happens everywhere for perfect substitutes and along the arms of the L for perfect complements. Not convex: the curve bows away from the origin, the chord dips below it, and the mix is worse than the endpoints.

Back to the snacks



One bundle: (weeks A, weeks B)

$(3, 0) \sim (0, 3)$ and convex $\succsim$
 $\Rightarrow$ every mix is $\succsim$ both

Week by week: no bundle in view

64 % mixed with the bundle in view, 9 % without. Convexity is a statement about **bundles**.

Simonson (1990), study 1. Two snacks A, B shown; the study offered six.

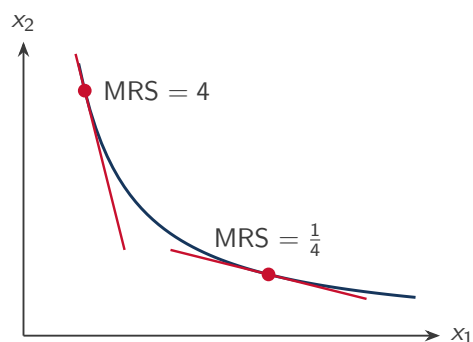
Slide 26 Back to the snacks

The snack study read through convexity. Treat the three weeks as one bundle (weeks of A, weeks of B) with $A + B = 3$, so the possible bundles are $(3, 0)$, $(2, 1)$, $(1, 2)$, $(0, 3)$. If a student is indifferent between three weeks of A and three weeks of B and has convex preferences, the mixes are at least as good as either extreme, because each is a weighted average of them: $(2, 1) = \frac{2}{3}(3, 0) + \frac{1}{3}(0, 3)$. Strictly better if preferences are strictly convex.

Choosing all at once, subjects saw the bundle and 64 % mixed. Choosing week by week, the bundle was never an object of choice, and 9 % mixed.

Takeaway: convexity is a property of preferences over bundles. Whether people frame three weeks as a bundle is an empirical question on which the theory is silent. The picture assumes $(3, 0) \sim (0, 3)$ for simplicity; if one extreme is strictly better, convexity only says the mixes are at least as good as the worse extreme.

The marginal rate of substitution is the slope



$MRS_{12}(x)$: good 2 given up per unit of good 1, staying indifferent
 $= -\text{slope of the curve at } x$

Monotone + convex: moving right along a curve, the MRS **never rises**; strictly convex: it falls.

With a utility function u : $MRS_{12} = (\partial u / \partial x_1) / (\partial u / \partial x_2)$, the quiz's last question.

Slide 27 The MRS is the slope

The marginal rate of substitution of good 1 for good 2 at x , $MRS_{12}(x)$, is how many units of good 2 the consumer gives up for one more unit of good 1 while staying indifferent. Geometrically it is minus the slope of the indifference curve at x .

Picture: the curve $x_1 x_2 = 4$. At $(1, 4)$ the tangent has slope -4 , so the MRS is 4: good 1 is scarce and valued highly. At $(4, 1)$ the slope is $-\frac{1}{4}$, so the MRS is $\frac{1}{4}$.

With monotone and convex preferences (and smooth curves) the MRS never rises as you move right along a curve; with strictly convex preferences it falls. "Diminishing MRS" is convexity in other words. The grey line previews the formula from the quiz's last question, the ratio of marginal utilities; it arrives with utility functions on Tuesday.

In pairs: five preferences on $\mathbb{R}_+^2$

	$x \succsim y \iff$	monotone	strongly	convex	strictly
perfect substitutes	$x_1 + x_2 \geq y_1 + y_2$				
perfect complements	$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$				
Cobb–Douglas type	$x_1 x_2 \geq y_1 y_2$				
quasilinear	$x_1 + \sqrt{x_2} \geq y_1 + \sqrt{y_2}$				
lexicographic	$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$				

Sketch the indifference set through $(2, 2)$ for each, then fill the row. Eight minutes.

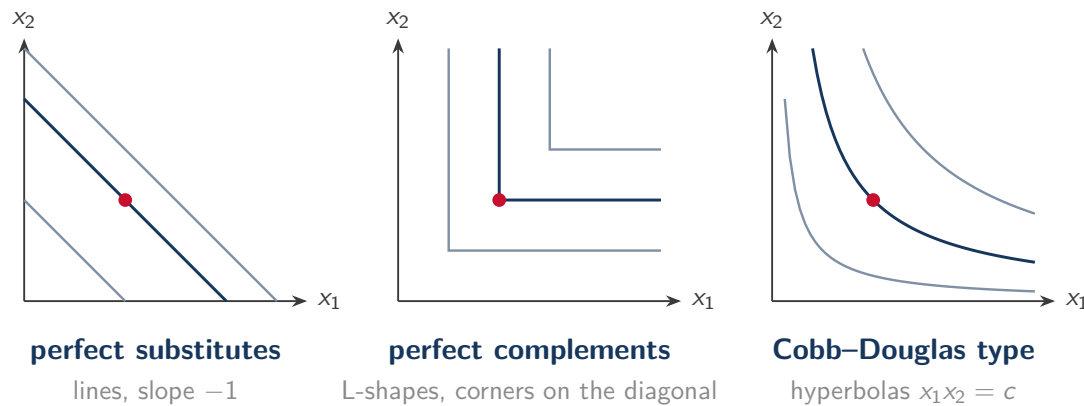
Slide 28 In pairs: five preferences

Five standard preferences on $\mathbb{R}_+^2$, axes included, each given by the rule for " x is at least as good as y ".

Try it: for each preference sketch the indifference set through $(2, 2)$, then decide whether it is monotone, strongly monotone, convex, strictly convex. The sketches matter more than the ticks: most properties can be read off the shape, and the crosses all need a counterexample.

Answers on the next three slides: the maps first, then the table with reasons.

Answers: the indifference maps, I



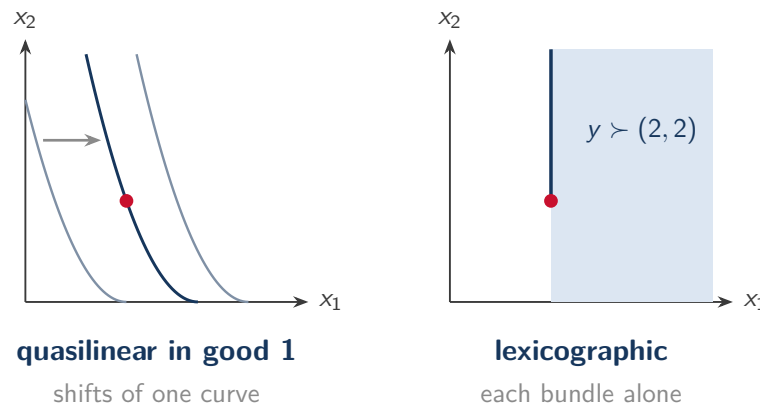
Red dot: $(2, 2)$. Pale curves: two other indifference sets of the same preference.

Slide 29 Answers: the indifference maps, I

Perfect substitutes: the indifference sets are parallel straight lines $x_1 + x_2 = c$ with slope -1 . Perfect complements: L-shapes with the corner on the 45-degree line; along each arm, extra units of one good alone add nothing. Cobb-Douglas type: hyperbolas $x_1 x_2 = c$ that approach the axes but never touch them.

In each panel the dark curve passes through $(2, 2)$ and the pale curves are two other indifference sets of the same preference.

Answers: the indifference maps, II



Quasilinear: $x_2 = (c - x_1)^2$, $x_1 \leq c$. Lexicographic: $\{y_1 > 2\}$ plus the half-line above $(2, 2)$.

Lexicographic indifference sets are single points; Tuesday: why that matters for utility (MWG 3.C).

Slide 30 Answers: the indifference maps, II

Quasilinear in good 1: the curve through $(2, 2)$ is $x_2 = (c - x_1)^2$ for $x_1 \leq c$, here with $c = 2 + \sqrt{2}$, and every other curve is a horizontal shift of it (the grey arrow); each curve meets the horizontal axis at $x_1 = c$.

Lexicographic: good 1 is compared first, and good 2 only breaks ties. The indifference set of $(2, 2)$ is the single point $(2, 2)$. The upper contour set is the open half-plane to the right, $y_1 > 2$, plus the half-line straight above $(2, 2)$.

Single points as indifference sets: remember this for Tuesday, where it explains why the lexicographic preference has no utility function.

Answers: five preferences on $\mathbb{R}_+^2$

	$x \succsim y \iff$	monotone	strongly	convex	strictly
perfect substitutes	$x_1 + x_2 \geq y_1 + y_2$	✓	✓	✓	✗
perfect complements	$\min\{x_1, x_2\} \geq \min\{y_1, y_2\}$	✓	✗	✓	✗
Cobb–Douglas type	$x_1 x_2 \geq y_1 y_2$	✓	✗ ⁺	✓	✗ ⁺
quasilinear	$x_1 + \sqrt{x_2} \geq y_1 + \sqrt{y_2}$	✓	✓	✓	✓
lexicographic	$x_1 > y_1$, or $x_1 = y_1$ and $x_2 \geq y_2$	✓	✓	✓	✓

Flat stretches kill strictness: substitutes, the arms of the L.

⁺ Both fail only on the axes: $(0, 1) \sim (0, 2)$, and the mix of $(1, 0)$ and $(2, 0)$ is $\sim (0, 0)$. Off the axes, both hold.

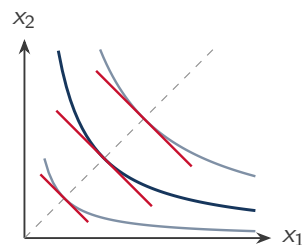
Slide 31 Answers: five preferences

Perfect substitutes are not strictly convex: the mix of two bundles on the same line stays on that line, so it is only indifferent. Perfect complements are not strongly monotone, $(1, 1) \sim (2, 1)$, and not strictly convex: $(2, 1)$ and $(3, 1)$ are both indifferent to $(1, 1)$, and so is their mix.

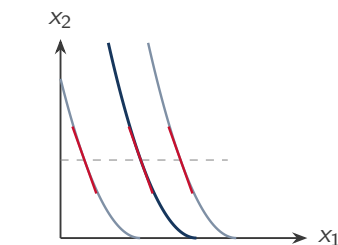
Cobb–Douglas fails both only on the axes: $(0, 1)$ and $(0, 2)$ both have product 0, so $(0, 1) \sim (0, 2)$ although $(0, 2)$ has more; and the mix of $(1, 0)$ and $(2, 0)$ is $(1.5, 0)$, indifferent to $(0, 0)$, not better. Off the axes both hold: always check the boundary of the consumption set.

Quasilinear $x_1 + \sqrt{x_2}$ is strongly monotone (raising either coordinate raises the number) and strictly convex, because the square root is strictly concave (if $y_2 = z_2$, the scores differ and the mix beats the lower one). Lexicographic is strictly convex too: if $y \neq z$ are both at least as good as x , either one has first coordinate above x_1 , and then so does the mix, or both sit on the vertical line through x , and then so does the mix, strictly above x .

Slide 32 Two families you will meet again



homothetic
same slope along a ray



quasilinear in good 1
same slope along a horizontal

homothetic: $x \sim y \Rightarrow \alpha x \sim \alpha y, \alpha \geq 0$

quasilinear in good 1: $x \sim y \Rightarrow x + \alpha e_1 \sim y + \alpha e_1, \alpha \in \mathbb{R}; x + \alpha e_1 \succ x, \alpha > 0$

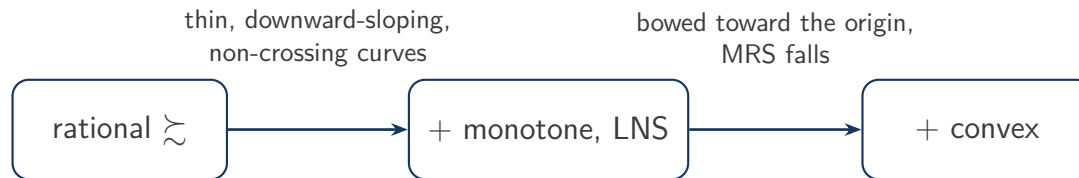
MWG Def. 3.B.6 (monotone $\succsim$ on $\mathbb{R}_+^2$), Def. 3.B.7 (on $X = \mathbb{R} \times \mathbb{R}_+$, so $\alpha < 0$ stays in X); $e_1 = (1, 0)$.

MWG Definitions 3.B.6 and 3.B.7; only 3.B.6 assumes a monotone preference. Homothetic: if $x \sim y$ then $\alpha x \sim \alpha y$ for every $\alpha \geq 0$. The whole indifference map is one curve scaled along rays from the origin, so along any ray the slope (the MRS) is the same: the red tangents on the 45-degree line are parallel. Cobb–Douglas, perfect substitutes and perfect complements are all homothetic.

Quasilinear with respect to good 1: if $x \sim y$ then $x + \alpha e_1 \sim y + \alpha e_1$ for every α , where $e_1 = (1, 0)$, and adding good 1 always makes a bundle strictly better. The curves are horizontal translates of each other, so along any horizontal line the slope is the same. MWG states this definition on $X = \mathbb{R} \times \mathbb{R}_+$, so that shifting left by α keeps the bundle in X .

Both families return with demand, in the income expansion paths of weeks 2 and 3.

Today in one line



Tuesday: when a number $u(x)$ can stand in for $\succsim$, and why the lexicographic one has none.
Problem set 1 goes out.

Read MWG 3.C (or Muñoz-Garcia ch. 1) for Tuesday 6 October.

Slide 33 Today in one line

Start with a rational $\succsim$. Add monotonicity or local non-satiation, and the indifference sets are thin, downward-sloping, non-crossing curves ordered outward. Add convexity, and the curves bow toward the origin, with a falling MRS.

Tuesday 6 October: utility representation (MWG 3.C). When can a number $u(x)$ stand in for $\succsim$, why the lexicographic preference has no such number, continuity, quasiconcavity. Problem set 1 goes out on Tuesday.

Reading for Tuesday: MWG 3.C, or Muñoz-Garcia chapter 1.